## SECTION-I

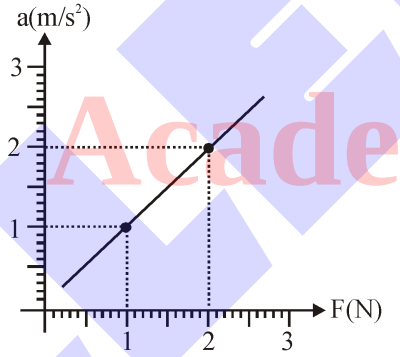
**Single Correct Answer Type** **19 Q. [3 M (–1)]**

**1.** A travelling microscope has a main scale graduated in $\frac{1}{2}$ mm.50 vernier divisions coincide with 49 main scale divisions. What will be the reading if the main scale reads 7.45 cm and the 29th division of the vernier coincides with the main scale division ?

(A) 7.459 cm (B) 7.453 cm

(C) 7.479 cm (D) 7.48 cm

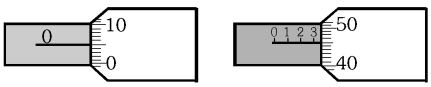
**2.** In an experiment to determine the inertial mass of an object using Newton's second law, following graph is obtained between net force on the object and the acceleration produced in it. The mass of the object within error limits is :-




(A) 1.0 kg (B) 1 kg

(C) (1.0 ± 0.1) kg (D) (1.0 ± 0.15)kg

**3.** The circular scale of a micrometer has 200 divisions and pitch of 2mm. Find the measured value of thickness of a thin sheet.




(A) 3.41 mm (B) 6.41 mm

(C) 3.46 mm (D) 3.51 mm

**4.** A screw gauge has a screw having 2 threads in 1 mm. The circular scale has 50 divisions. Find the diameter of wire, if the main scale shows 6th division and the vernier reads 46.

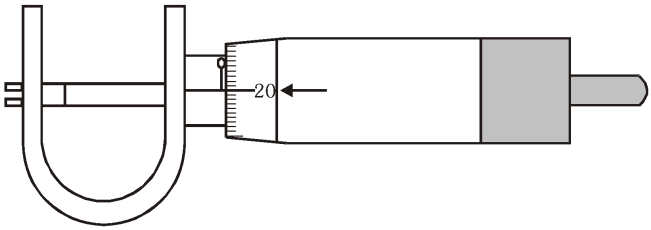
(A) 6.46 mm (B) 3.46 mm

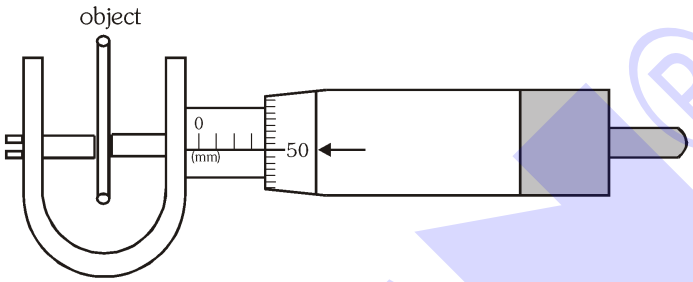
(C) 6.54 mm (D) 3.04 mm

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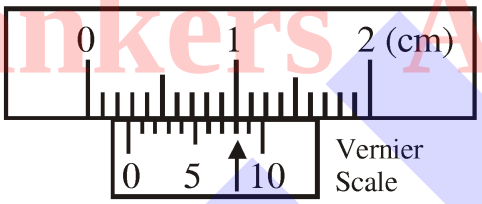

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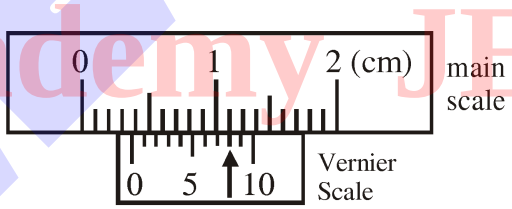
**5.** When a piece of wire is held diametrically in a screw gauge [pitch = 1mm, number of division on circular scale = 100]. The readings obtained are as shown:

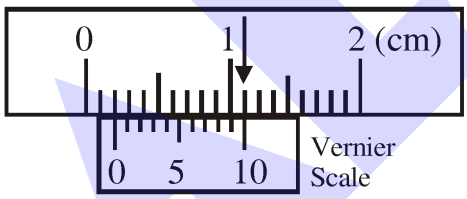


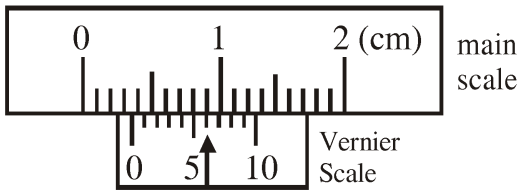




Now if we measure the same with help of vernier callipers [1 MSD = 1mm, 10 divisions of vernier coinciding with 9 divisions of main scale] having a negative zero error of 0.5 mm, then find which of the following figures correctly represents the reading.

(A) 


(B) 


(C) 


(D) 


**6.** A student obtained following observations in an experiment of meter bridge to find the unknown resistance of the circuit :

| S.No. | R | $\ell$ | 100 – $\ell$ | $S=\left(\frac{100-\ell}{\ell}\right)R$ |
|---|---|---|---|---|
| 1 | 20Ω | 43 | 57 | 26.51 Ω |
| 2 | 30Ω | 51 | 49 | 28.82 Ω |
| 3 | 40Ω | 59 | 41 | 27.80 Ω |
| 4 | 60Ω | 70 | 30 | 25.71 Ω |

The most accurate value of unknown resistance is :-

(A) 26.51 Ω (B) 28.82 Ω (C) 27.80 Ω (D) 25.71 Ω

**7.** Relative density of the material of a body is found by weighing it first in air and then in water. The weight in air is (5.00 $\pm$ 0.05) newton and weight in water is (4.00 $\pm$ 0.05) newton. Then the relative density of material should be:-

(A) 5.0 $\pm$ 11% (B) 5.0 $\pm$ 1% (C) 5.0 $\pm$ 6% (D) 1.25 $\pm$ 5%

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**8.** The viscosity of a fluid μ, can be determined by measuring the terminal velocity $V_T$ of a sphere when it descends in the fluid. The fluid has a density $\rho_f$ while the sphere has a density $\rho_s$ and a diameter of d. The viscosity can then be calculated by the formula

$$\mu = \frac{5(\rho_s - \rho_f)}{9V_T} d^2$$

The values measured are
$V_T = (1.60 \pm 0.04)\ ms^{-1}$
$\rho_s = (2700 \pm 20)\ kg\ m^{-3}$
$\rho_f = (900 \pm 10)\ kg\ m^{-3}$
$d = (20.0 \pm 0.4)\ mm$
What is the percentage uncertainty in the value of μ?
(A) 6.2% (B) 7.1% (C) 8.2% (D) 8.4%

**9.** In a screw gauge, there are 100 division in circular scale and each main scale division is of 1 mm. When there is no gap between the jaws, 97$^{th}$ division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3 mm mark and 4 mm mark such that the 76$^{th}$ division of circular scale coincides with the reference line. Select the correct alternative :
(A) The least count of the micrometer is 0.01 cm
(B) The zero error is –0.04 mm
(C) The diameter of the ball is 3.79 mm
(D) There is positive zero error in the screw guage

**10.** Three students make measurements of the length of a 1.50 meter rod. Each student reports an uncertainty estimate representing an independent random error applicable to the measurement.
Alice : A single measurement using a 2.0 meter long tape measure, to within ±2 mm.
Bob : Two measurements using a wooden meter stick, each to within ±2 mm, which he adds together.
Christina : Two measurements using a machinist's meter ruler, each to within ±1 mm, which she adds together.
The students' teacher prefers measurements that are likely to have less error. Which is the correct order of preference?
(A) Christina's is preferable to Alice's, which is preferable to Bob's
(B) Alice's is preferable to Christina's, which is preferable to Bob's
(C) Alice's and Christina's are equally preferable; both are preferable to Bob's
(D) Christina's is preferable to both Alice's and to Bob's, which are equally preferable

**11.** In an experiment to study standing waves, you use a string whose mass per length is $\mu = (1.0 \pm 0.1) \times 10^{-4}$ kg/m. You look at the fundamental mode, whose frequency f is related to the length L and tension T of the string by the following equation $L = \frac{1}{2f}\sqrt{\frac{T}{\mu}}$. You make a plot with L on the y-axis and $\sqrt{T}$ on the x-axis, and find that the best fitting line is $y = (8.0 \pm 0.3) \times 10^{-3}\, x + (0.2 \pm 0.04)$ in SI units. What is the value of the frequency of the wave (including the error)? Express your result in SI unit (Hz).
(A) 6250 ± 859.8 Hz (B) 1250 ± 32.3 Hz (C) 6250 ± 546.9 Hz (D) 4875 ± 287 Hz

**12.** You measure two quantities as $A = 1.0\ m \pm 0.2\ m$, $B = 2.0\ m \pm 0.2\ m$. We should report correct value for $\sqrt{AB}$ as:
(A) 1.4 m ± 0.4 m (B) 1.41m ± 0.15 m (C) 1.4m ± 0.3 m (D) 1.4m ± 0.2 m


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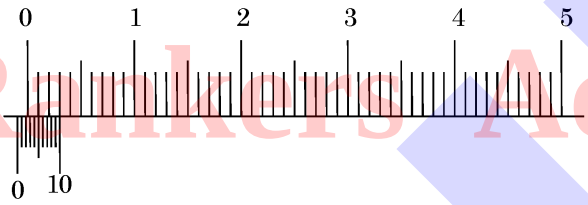
**13.** In a meter bridge experiment the resistance of resistance box is 16Ω, which is inserted in right gap.The null point is obtained at 36cm from the left end. The least count of meter scale is 1mm. What is the value of unknown resistance? (Error = L.C. or L.C./2)

(A) $9 \pm \frac{5}{128}\Omega$ (B) $9 \pm \frac{5}{256}\Omega$

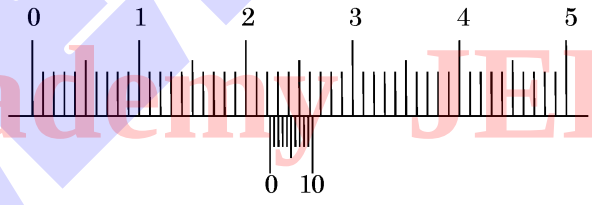
(C) $9 \pm \frac{5}{512}\Omega$ (D) $\frac{128}{9} \pm \frac{1}{2560}\Omega$

**14.** In a specially designed vernier calipers, 10 divisions of the vernier scale are equal to 4 divisions of the main scale. In the following figure, the zero error and a measurement are shown. In the zero error, the 5th and the 10th divisions of the vernier coincides with 1 mm and 3 mm mark on the main scale respectively and in the measurement, the 2nd and the 7th divisions of the vernier scale coincide with the 23 mm and 25 mm mark on the main scale respectively as shown in the figure. Report the length measured with due regards to the error.

Zero Error




Reading of Measurement




(A) $(21.2 \pm 0.2)$ mm (B) $(23.2 \pm 0.2)$ mm

(C) $(23.8 \pm 0.4)$ mm (D) $(23.4 \pm 0.6)$ mm

**15.** A student performs an experiment to determine the Young's modulus of a wire, exactly 2m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of $\pm$ 0.04 mm at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of $\pm 0.01$ mm. Take $g = 10\,m/s^2$ (exact). The Young's modulus obtained from the reading is

(A) $(2.0 \pm 0.3) \times 10^{11}\ N/m^2$ (B) $(2.0 \pm 0.2) \times 10^{11}\ N/m^2$

(C) $(2.0 \pm 0.1) \times 10^{11}\ N/m^2$ (D) $(2.0 \pm 0.05) \times 10^{11}\ N/m^2$

**16.** Rank the vernier callipers/screw gauge in the order of their increasing precision :

(1) Screw gauge having main scale division 1mm and no. of circular division = 200.

(2) Screw gauge having main scale division 0.5 mm and no. of circular division = 400.

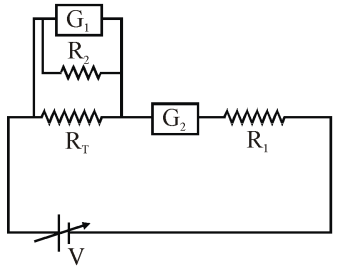
(3) Vernier callipers having main scale division = 1mm such that 9 MSD = 10 VSD.

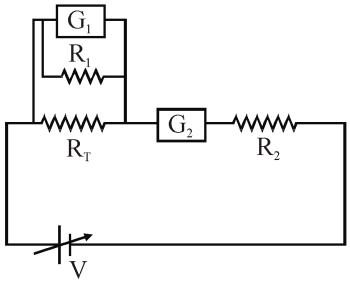
(4) Vernier callipers having main scale division = 1mm such that 4 MSD = 5 VSD.

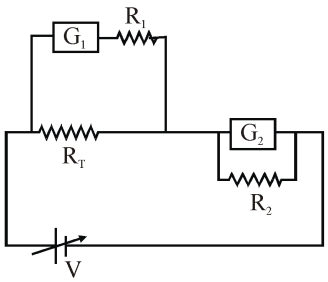
(A) 4 < 2 < 3 < 1 (B) 2 < 4 < 1 < 3 (C) 4 < 3 < 2 < 1 (D) 4 < 3 < 1 < 2

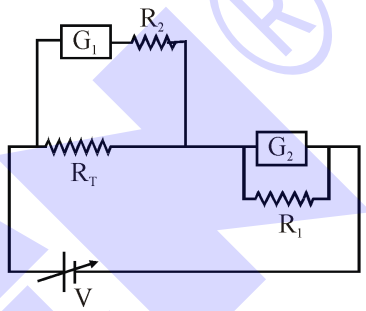

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**17.** To verify Ohm's law, a student is provided with a test resistor $R_T$, a high resistance $R_1$, a small resistance $R_2$, two identical galvanometers $G_1$ and $G_2$, and a variable voltage source V. The correct circuit to carry out the experiment is :– **[JEE 2010]**

(A) 


(B) 


(C) 


(D) 


**PARAGRAPH "A"**

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation z = x/y. If the errors in x, y and z are Δx, Δy and Δz, respectively, then **[JEE-Advance 2018]**

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in Δy/y, is $1 \mp (\Delta y / y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1, \frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

**(There are two questions based on Paragraph "A", the question given below is one of them)**

**18.** Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a. If the error in the measurement of a is Δa(Δa/a << 1), then what is the error Δr in determining r ?

(A) $\frac{\Delta a}{(1+a)^2}$ (B) $\frac{2\Delta a}{(1+a)^2}$ (C) $\frac{2\Delta a}{(1-a^2)}$ (D) $\frac{2a\Delta a}{(1-a^2)}$


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**PARAGRAPH "A"**

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation z = x/y. If the errors in x, y and z are Δx, Δy and Δz, respectively, then

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in Δy/y, is $1 \mp (\Delta y / y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1, \frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

**(There are two questions based on Paragraph "A", the question given below is one of them)**

**19.** In an experiment the initial number of radioactive nuclei is 3000. It is found that 1000 ± 40 nuclei decayed in the first 1.0 s. For $|x| << 1$, In (1 + x) = x up to first power in x. The error Δλ, in the determination of the decay constant λ, in $s^{-1}$, is :-

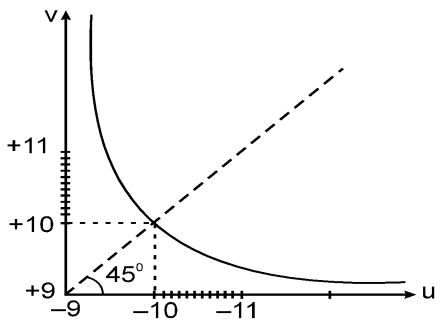
(A) 0.04 (B) 0.03 (C) 0.02 (D) 0.01

**Multiple Correct Answer Type** **5 Q. [4 M (–1)]**

**20.** A teacher asks 4 students to find the speed of sound using resonance column apparatus. The set of apparatus used by each of student was different. The frequency of tuning fork used was known precisely. He wants the error in speed of sound to be within 1%. Which of the student(s) did the experiment correctly. Their results are tabulated below.

| **Student** | **Frequency** | **1st resonance** | **2nd resonance** |
|---|---|---|---|
| (A) | 476 Hz | 17.0 cm | 50.0 cm |
| (B) | 340 Hz | 25.2 cm | 76.0 cm |
| (C) | 544 Hz | 15.6 cm | 46.0 cm |
| (D) | 510 Hz | 16.0 cm | 48.2 cm |

**21.** The graph between object distance u and image distance v for a lens is given below. The value of u and v are in centimeter.Then mark the correct statement(s).



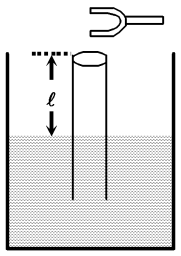

(A) The focal length of the lens is 5cm. (B) The focal length of the lens is 0.5cm.

(C) The error in focal length is 0.05 cm (D) The fractional error in focal length is 0.1


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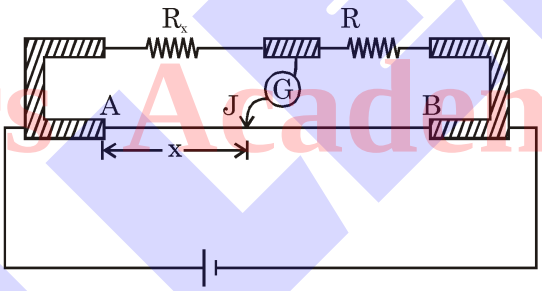
**22.** A flat circular annular disc has a mass M, outer radius a and inner radius b. The measured values of M, a and b are, M = (0.300 kg ± 0.003 kg), a = (4 cm ± 0.01 cm) & b = (2 cm ± 0.01 cm) :-
(A) Percentage error in the calculation of a and b are same.
(B) Percentage error in the calculation of M is 1%
(C) Percentage error in the calculation of moment of inertia of the disc about an axis passing through centre and perpendicular to plane of disc is 1.6 %.
(D) Percentage error in the calculation of moment of inertia of the disc about an axis passing through centre and perpendicular to plane of disc is 3.2 %.

**23.** A tuning fork of known frequency is held at the open end of a long tube, which is dipped into water as shown in the figure. The tuning fork of frequency 165 Hz resonates in first and second harmonic when air columns have lengths $\ell_1 = 50 \pm 0.5$ cm and $\ell_2 = 150 \pm 0.1$ cm. Then




(A) Speed of sound in air is (330 ± 1.98) m/s
(B) Speed of sound in air is (330 ± 1.2) m/s
(C) Percentage maximum error in speed of sound is 0.6%
(D) Percentage maximum error in speed of sound is 0.4%

**24.** Figure shows an experimental setup to find the value of an unknown resistance $R_X$ using a meter bridge. AB is uniform meter bridge wire of length L = 100 cm. When sliding jockey is at J (AJ = x), the galvanometer shows zero deflection. Length AJ is measured by a meter scale having leastcount of 1mm. In an experiment known resistance is taken as 20 Ω length was found to be (20.0 ± 0.1)cm




(A) Value of $R_x = (5.00 \pm 0.03)\Omega$
(B) Value of $R_x = (5.00 \pm 0.02)\Omega$
(C) To minimize fractional error the known resistance should be very close to $R_X$
(D) When fractional error is minimized $x = \frac{L}{2}$

## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** **5 Q. [4 M (0)]**

**25.** Volume of a gas undergoing certain process varis with temperature as shown $V = V_0\left(\frac{T_0}{T}\right)^{1/2} e^{\left[\frac{2(T-T_0)}{3\ T_0}\right]}$ where $T_0$ is initial temperature, $V_0$ is initial volume. If apparatus rise for measuring temperature can measure temperature with an accuracy of 0.2 k, find the percentage error in calculation of volume at temp $T = \frac{3T_0}{4}$ ($V_0$ and $T_0$ are accurately known)

**26.** The force exerted on a length *L* of a wire carrying a current *I* by a magnetic field perpendicular to the wire is measured in an experiment and a graph is prepared between modulus *F* of the force and segment length *L*. The plot is a straight line, with slope $s = (10 \pm 1) \times 10^{-5}$ A·T. The current in the wire is $I = (15 \pm 1)$ mA. The modulus of the magnetic induction in SI units, including the error is found to be $(x \pm y) \times 10^{-3}$ T. Find value of x/y.


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**27.** An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured values of length $\ell$(in cm) of the pipe and radius r (in cm) of the pipe are $\ell = 94 \pm 0.1$, $r = 5 \pm 0.05$. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by this experiment is given by $\alpha^2\%$. Find the value of $10\alpha$.

**28.** A hemispherical bowl of radius R = 0.1m is rotating about its own axis (which is vertical) with an angular velocity ω. A particle of mass $10^{-2}$ kg on the frictionless inner surface of the bowl is also rotating with the same ω. The particle is at a height h from the bottom of the bowl. It is desired to measure 'g' using this set up, by measuring h accurately. Assuming that R and ω are known precisely and that the least-count in the measurement of h is $10^{-4}$ m, the numerical value of minimum possible error Δg in the measured value of g is given by $\alpha \times 10^{-2}$ m/s$^2$. Fill the value of α in OMR sheet.

**29.** The diameter of a wire of length 100 cm is measured with the help of a screw gauge. The main scale reading is 1 mm and circular scale is reading is 25. Pitch of the screw gauge is 1 mm and the total number of divisions on the circular scale is 100. This wire is used in an experiment of determination of Young's modulus of a wire by Searle's method. The following data are available: elongation in the wire $l = 0.125$ cm under the tension of 50 N, least count for measuring normal length of wire is 0.01 cm and for elongation in the wire is 0.001 cm. The maximum error in the calculating value of Young's modulus (Y), assuming that the force is measured very accurately, is $\frac{8n}{10}\%$, where n is very nearest interger. Find the value of n.

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**GR # ERROR** — **ANSWER KEY**

## SECTION-I

**Single Correct Answer Type** — **19 Q. [3 M (–1)]**

| | | | | |
|---|---|---|---|---|
| 1. Ans. (C) | 2. Ans. (C) | 3. Ans. (B) | 4. Ans. (B) | 5. Ans. (A) |
| 6. Ans. (B) | 7. Ans. (A) | 8. Ans. (C) | 9. Ans. (C) | 10. Ans. (C) |
| 11. Ans. (C) | 12. Ans. (D) | 13. Ans. (B) | 14. Ans. (B) | 15. Ans. (B) |
| 16. Ans. (D) | 17. Ans. (C) | 18. Ans. (B) | 19. Ans. (C) | |

**Multiple Correct Answer Type** — **5 Q. [4 M (–1)]**

| | | | | |
|---|---|---|---|---|
| 20. Ans. (B,D) | 21. Ans. (A, C) | 22. Ans. (B, C) | 23. Ans. (A, C) | 24. Ans. (A,C,D) |

## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** — **5 Q. [4 M (0)]**

| | | | | |
|---|---|---|---|---|
| 25. Ans. 0 | 26. Ans. 7 | 27. Ans. 4 | 28. Ans. 1 | 29. Ans. 3 |


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